6 A soccer ball of mass 0.45 kg is travelling towards a player at 3.2 m s^{-1}.

The player kicks the ball with an impulse of 6.7 N s, returning it in the direction of approach.
What is the new speed of the ball?

A 5.3 m s^{-1} **B** 8.1 m s^{-1} **C** 12 m s^{-1} **D** 18 m s^{-1}